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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Hackstreet Boys
Selected Domain	Domain 2 : Orbyte Smart SME Challenge (Less Busywork, Better Business)
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth
Project / Solution Name	Stock Manager
Name (Leader)	Josiah Chong Ren Shun
Email & Phone (Leader)	josiahc159@gmail.com (+60 16-843 4553)
University Affiliation(s)	Xiamen University Malaysia, Monash University Malaysia
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small shops (sundry shops or kedai runcit, minimarts, small retail and F&B outlets) still track stock in notebooks, spreadsheets and WhatsApp. The number in the record rarely matches what is on the shelf. Damage, expiry, theft, miscounts and unrecorded sales go unnoticed until a customer asks for an item the records say is in stock, or until the owner spends an evening recounting everything.

The problem exists even in large, well-run retailers: a study of nearly 370,000 inventory records across 37 stores found 65% were inaccurate. US retailers lost an average of 1.6% of sales to shrink in FY2022. Small shops have fewer controls, yet cannot afford enterprise inventory systems or the training they require.

Owners lose money and evenings. Staff are blamed when numbers do not match, because there is no record of who changed what. When a clerk quietly "fixes" a count, nobody approves it and nothing is logged, so mistakes and theft stay hidden.

This matters economically. MSMEs produced 39.5% of Malaysia's GDP and 48.7% of its employment in 2024. Every ringgit lost to untracked stock is margin a micro-business cannot spare. Reliable stock records are also the starting point for reordering on time, planning cash flow and adopting connected tools such as Orbyte, the technology-driven productivity upgrade that SDG Target 8.2 calls for.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Stock Manager is a web app for the Orbyte hub that turns a shelf count into a trusted, approved stock record. It runs one complete automated journey:

1. Trigger: a clerk enters what is physically on the shelf.
2. Connected data: the app pulls the product's system quantity, unit price and stock-movement history.
3. Decision: a rule engine calculates the difference and its ringgit value. Differences of 2 units or fewer and under RM50 are corrected automatically. Larger, negative or outdated counts go to the owner, with the reason shown in plain words.

4. Human approval: the demo interface shows approval controls only in the Owner view. Production authentication and server-side role enforcement are planned. A count made before a later stock change cannot be approved.
5. Visible outcome: stock updates, every action is recorded in the current SQLite database, and approved changes plus low-stock alerts are included in a simulated Orbyte payload.

Where AI fits:

Following the Domain 2 brief to use AI only where it adds value, the approval decision itself is deliberately rule-based, because owners must be able to predict and audit why stock changed. For Phase 2 we plan to add AI where judgement helps: (a) pattern detection over the count and audit history to flag products or shifts with recurring losses before they grow; and (b) a GenAI-written weekly summary that explains, in Bahasa Melayu or English, what went missing, its value and what to check. AI will only recommend; the owner still decides.

Workshop skills used:

Database architecture: a relational SQLite schema built with SQLAlchemy, where stock, movement and audit records are written in one transaction. Cloud hosting: deployment on Replit for a public live prototype. APIs: a documented FastAPI REST API consumed by a React frontend, plus a simulated Orbyte sync endpoint. GenAI APIs: planned for the Phase 2 summary.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Technology	What it does
Frontend	React 19 + Vite 6	Dashboard, products, stock count, approvals and activity log screens; Owner / Stock clerk role switch (simulated login)
Backend API	Python, FastAPI, Pydantic	REST endpoints, input validation, OpenAPI / Swagger documentation
Database	SQLite + SQLAlchemy 2	Products, stock movements, adjustment requests and audit log, written in single transactions
Decision engine	Python rule service	Auto-fix when the difference is 2 units or fewer and under RM50; otherwise route to the owner; block outdated or negative counts
Import	Openpyxl, python-multipart	CSV / XLSX product import with a validation preview
Orbyte integration	Simulated sync endpoint	Packages approved changes and low-stock alerts as an Orbyte-shaped JSON payload and logs it; no external calls
Testing	Pytest, httpx	Workflow, risk-boundary, stale-approval and rollback tests

Hosting & Code	Replit, GitHub	Public live prototype; version history

The clerk and owner use the React frontend; the FastAPI backend applies the approval rules and writes stock and audit records to SQLite in one transaction. Approved changes and low-stock alerts are included in a simulated Orbyte payload; no external request is sent.

Sponsor-provided GenAI API or cloud platform: Yes. Replit (sponsor-provided) hosts the live prototype. RunPod is not used, because the Phase 1 MVP has no GPU workload. No GenAI API is used in Phase 1.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we studied how Malaysian micro-shops keep stock (notebooks, spreadsheets, WhatsApp) and defined two personas: the owner, who carries the losses and approves changes, and the clerk, who counts shelves but should not be able to change stock silently.

Define: one problem, untrusted stock records, rather than a broad "inventory system".

Ideate: we compared payment chasing, food-waste alerts and count reconciliation, and chose reconciliation because it is frequent, measurable and needs human approval.

Prototype: a working MVP of the full journey, built on synthetic data from a Malaysian minimarket (Milo, Maggi, cooking oil, rice, sardines).

Test: automated tests cover the risk boundaries, negative counts, outdated approvals and rollback; a scripted walkthrough checks each screen as both roles.

User-friendliness comes from previewing the decision before submitting, explaining every decision in plain words, one-tap reason buttons and a phone-friendly layout, so no formal training is needed.

Deployment readiness: hosted on Replit with one-command seed data and documented APIs. In Phase 2 we will test with 3 to 5 shop owners and measure time per count.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.2: "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors." Supported by Target 8.3: encourage the formalisation and growth of micro-, small- and medium-sized enterprises.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Public benchmark: 65% of inventory records were inaccurate across about 370,000 records in 37 stores; US retailers reported average shrink of 1.6% of sales in FY2022. For our simulated manual-workflow baseline, we assume stock corrections have no structured named approval or audit record. We will measure the pilot's own starting mismatch rate before testing.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Inventory record accuracy: % of products whose system quantity matches the physical count. Supporting KPIs: % of stock adjustments with a named approver or rule and an audit entry; number of items the owner must review per full count.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	After measuring the simulated pilot's starting mismatch rate across 5 to 20 products, target a 20% relative reduction after two count cycles. Target 100% of stock adjustments with an audit entry and a named approver or recorded rule. Test whether the auto-fix rules reduce owner reviews by at least 50%; treat this as a hypothesis, not a guaranteed outcome.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

Sources used in Sections 2 and 4

1. DeHoratius, N. and Raman, A. (2008), *Inventory Record Inaccuracy: An Empirical Analysis*, Management Science 54(4), 627-641. https://doi.org/10.1287/mnsc.1070.0789

2. National Retail Federation (2023), *National Retail Security Survey 2023*. https://nrf.com/research/national-retail-security-survey-2023

3. Department of Statistics Malaysia (2025), *Micro, Small & Medium Enterprises (MSMEs) Performance 2024*. https://www.dosm.gov.my/portal-main/release-content/micro-small--medium-enterprises-msmes-performance-2024

4. United Nations, *Sustainable Development Goal 8*, Targets 8.2 and 8.3. https://sdgs.un.org/goals/goal8

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
FastAPI	**Code (backend framework)**	**MIT**
Uvicorn	**Code (web server)**	**BSD-3-Clause**
SQLAlchemy	**Code (database toolkit)**	**MIT**
SQLite	**Code (database engine)**	**Public domain**
Pydantic	**Code (data validation)**	**MIT**
Python-multipart	**Code (file uploads)**	**Apache-2.0**
Openpyxl	**Code (Excel import)**	**MIT**
Pytest, httpx	**Code (Testing)**	**MIT**
React, React DOM	**Code (Frontend Library)**	**MIT**

Vite	**Code (frontend build tool)**	**MIT**

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

October 24th 2026